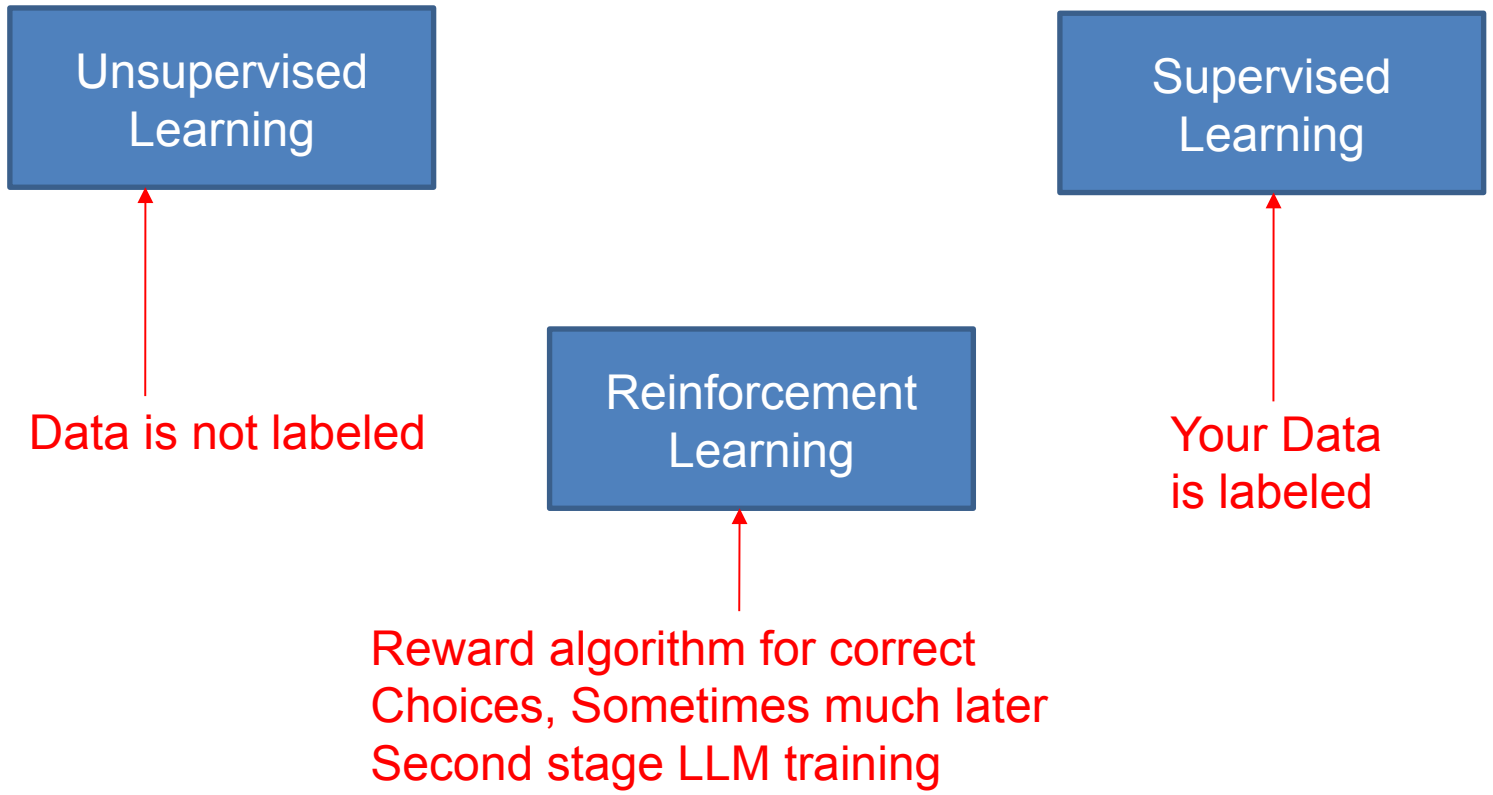
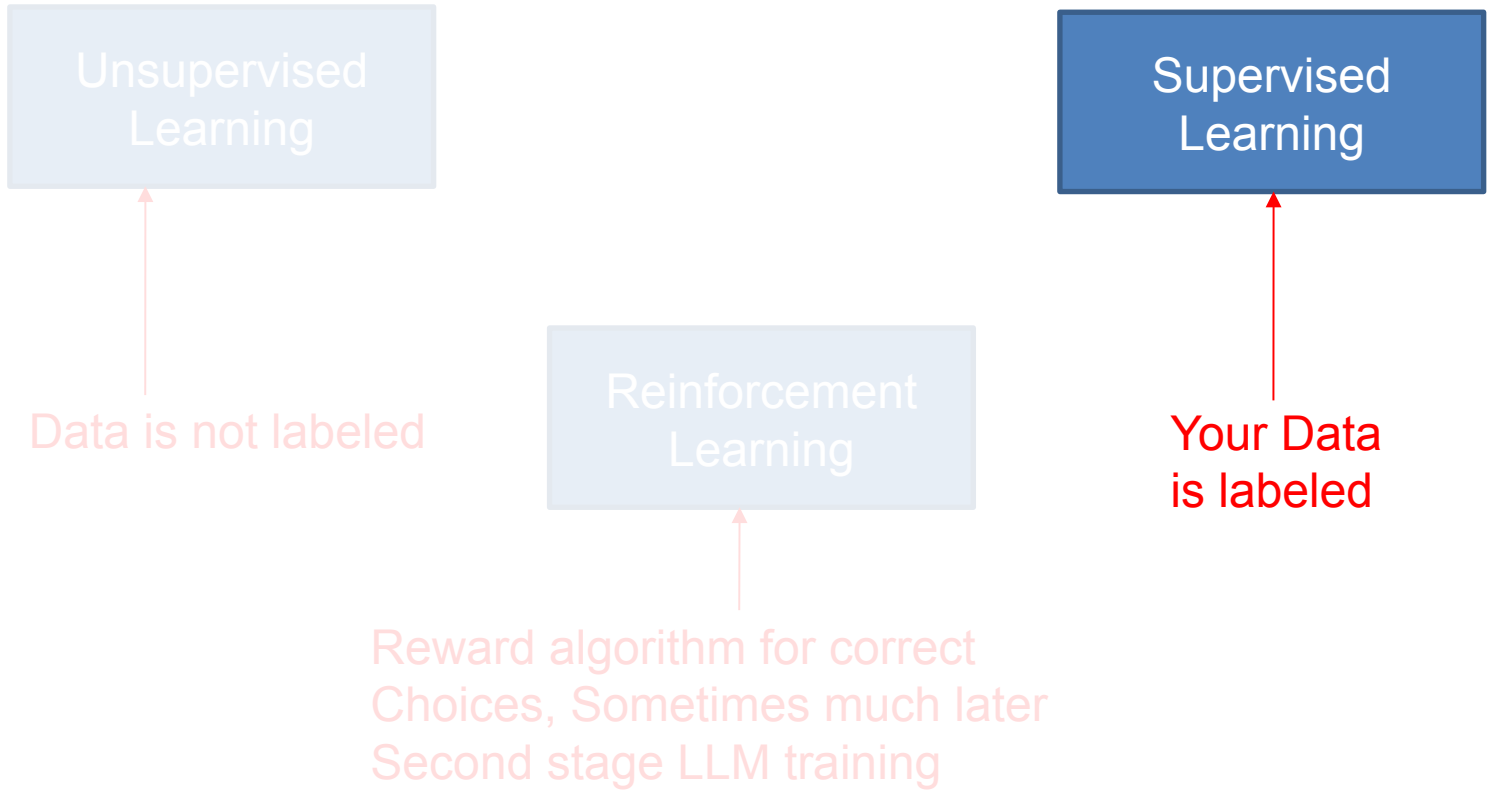
# Supervised Learning
# Splitting the dataset

# Some types of machine learning



# Some types of machine learning



# Aside: Semi-Supervised Learning

**Have a little labeled data, and a lot of unlabeled data**
**Train a classifier on the labeled data**
**Use that classifier to predict what unlabeled data is**
**Add all high confidence (>80%)* predictions to labeled dataset**

https://towardsdatascience.com/dimensionality-reduction-for-data-visualization-pca-vs-tsne-vs-umap-be4aa7b1cb29

# Supervised Learning

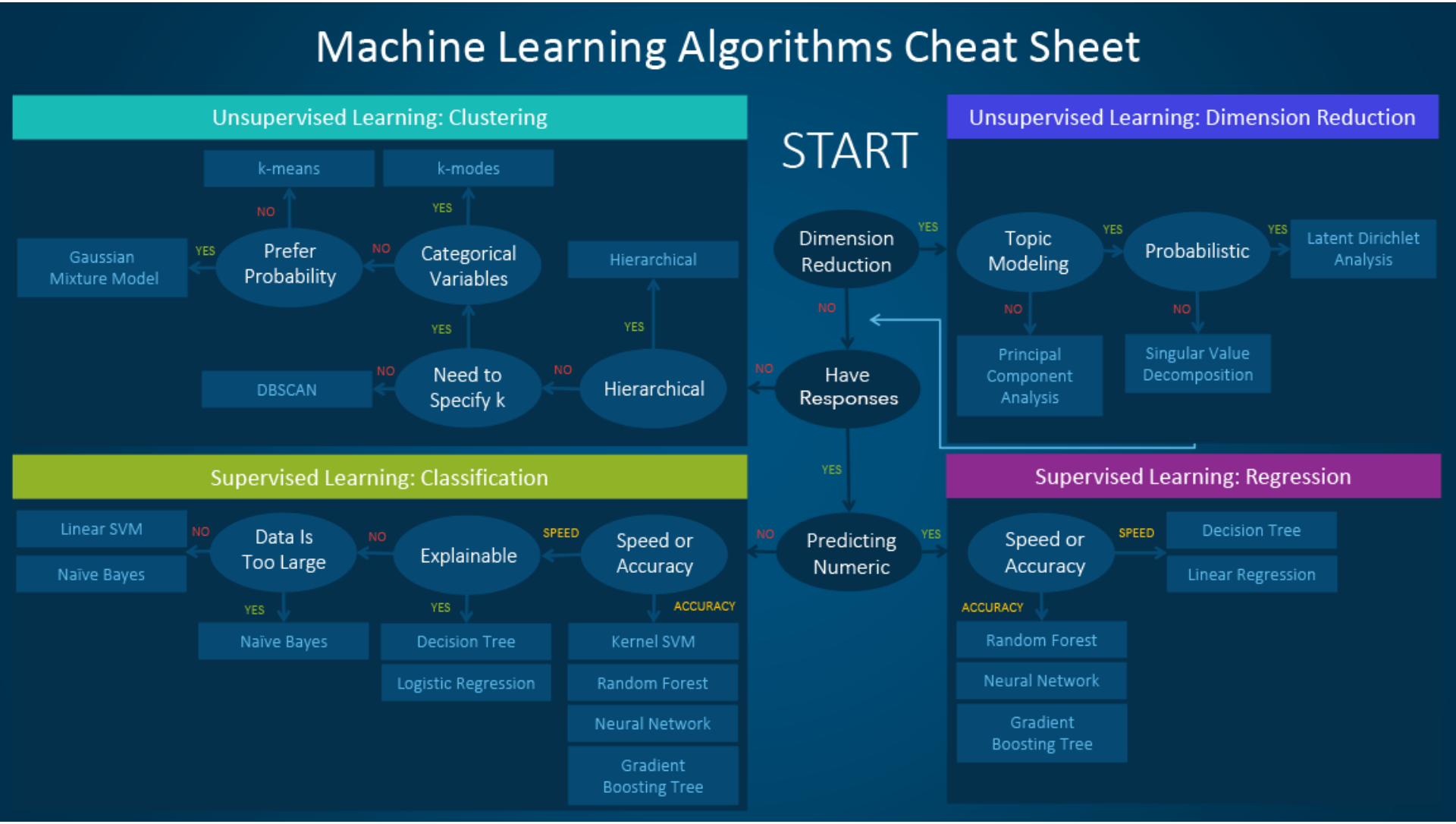
| | sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | target |
|---|---|---|---|---|---|
| 0 | 5.1 | 3.5 | 1.4 | 0.2 | setosa |
| 1 | 4.9 | 3.0 | 1.4 | 0.2 | setosa |
| 2 | 4.7 | 3.2 | 1.3 | 0.2 | setosa |
| 3 | 4.6 | 3.1 | 1.5 | 0.2 | setosa |
| 4 | 5.0 | 3.6 | 1.4 | 0.2 | setosa |

Known Target

**Supervised**

- Labeled Data
- Model knows correct result during training
- Uses correct result to adjust model params during training so its more likely to pick correct result
- Goal is to train model to pick correct result on unseen data
- Examples: Regressions, Decision Trees, Neural Networks

# Supervised and Unsupervised Learning




From https://blogs.sas.com/content/subconsciousmusings/2020/12/09/machine-learning-algorithm-use/

# Data

https://towardsdatascience.com/dimensionality-reduction-for-data-visualization-pca-vs-tsne-vs-umap-be4aa7b1cb29

# Data: Supervised Setup (easy part)

Still need to do pre-processing (standardization for neural networks)

Additionally:

- Need to divide data into 3 portions;  train, validation, test
- What about unbalanced datasets?
- Typically 80% train 20% test (or 70-30, or 90-10 depends on amount of data you have)
- Calculate standardization metrics on the train set only. Apply these IDENTICAL metrics to the test set.
- If you violate this rule, test set information will leak over to the train set.

# Data: Training a Model

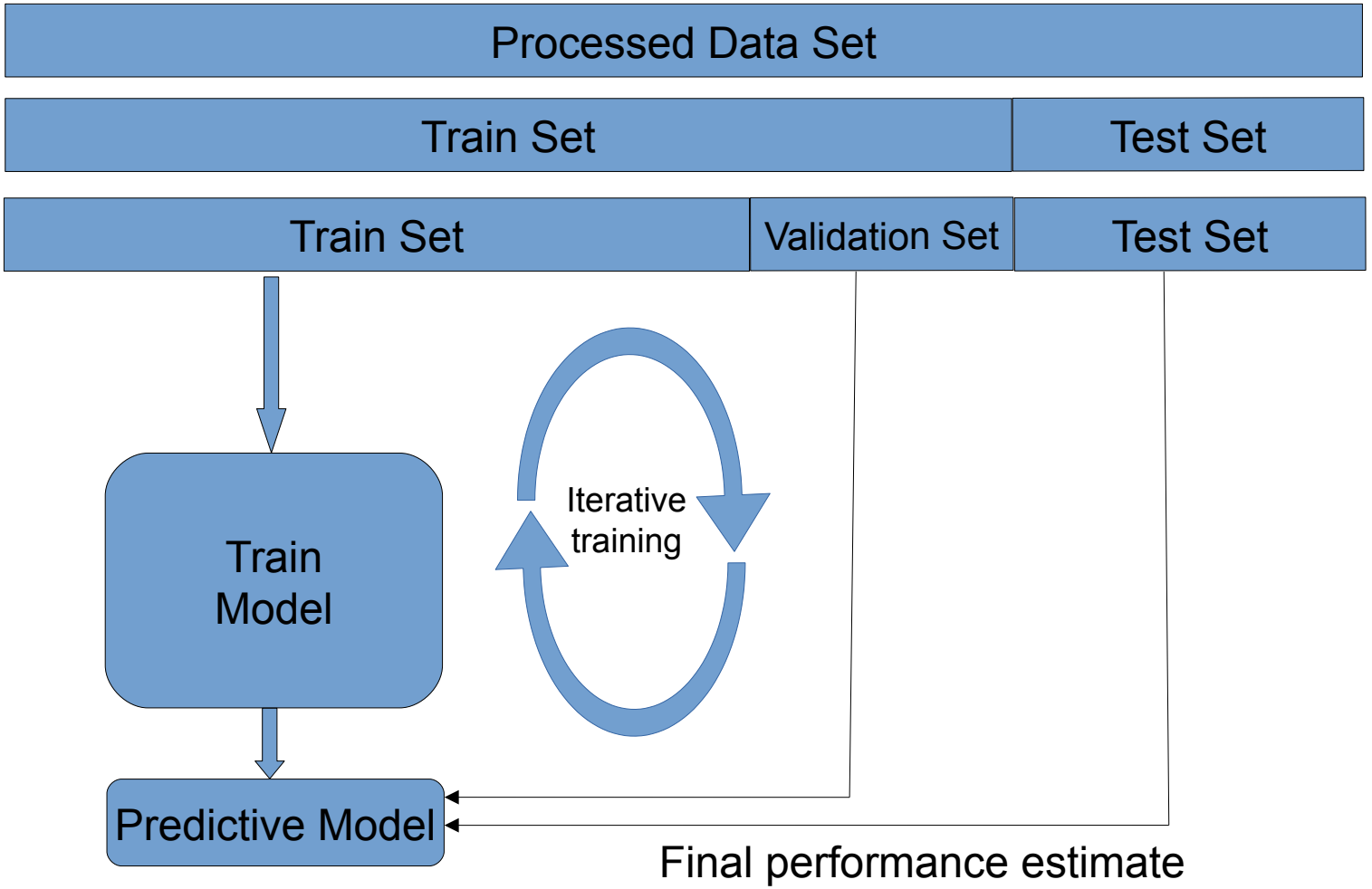



Figure from 'Python Machine Learning' by Sebastian

# Data: Balanced datasets

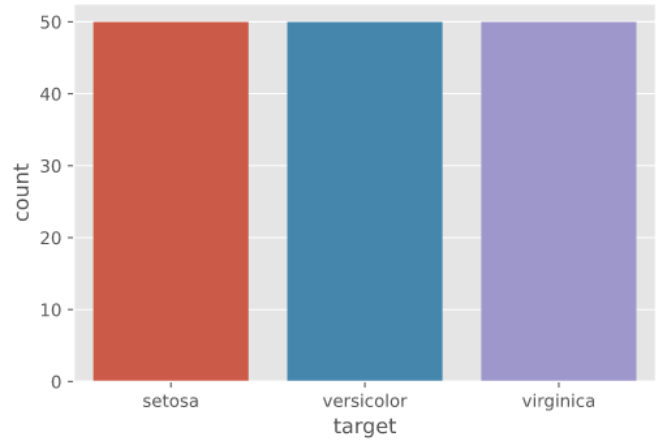
If your dataset has roughly the same number of labeled classes then its easy to do a train test split. Here we load the iris dataset, 77% of it is used to train, 33% to test.

```
from sklearn.model_selection import train_test_split
from sklearn import datasets
iris = datasets.load_iris()

X, y = iris.data, iris.target
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.33,random_state=42)
```

Notice we have an equal number of iris types

```
ax=sns.countplot(x="target", data=df)
```



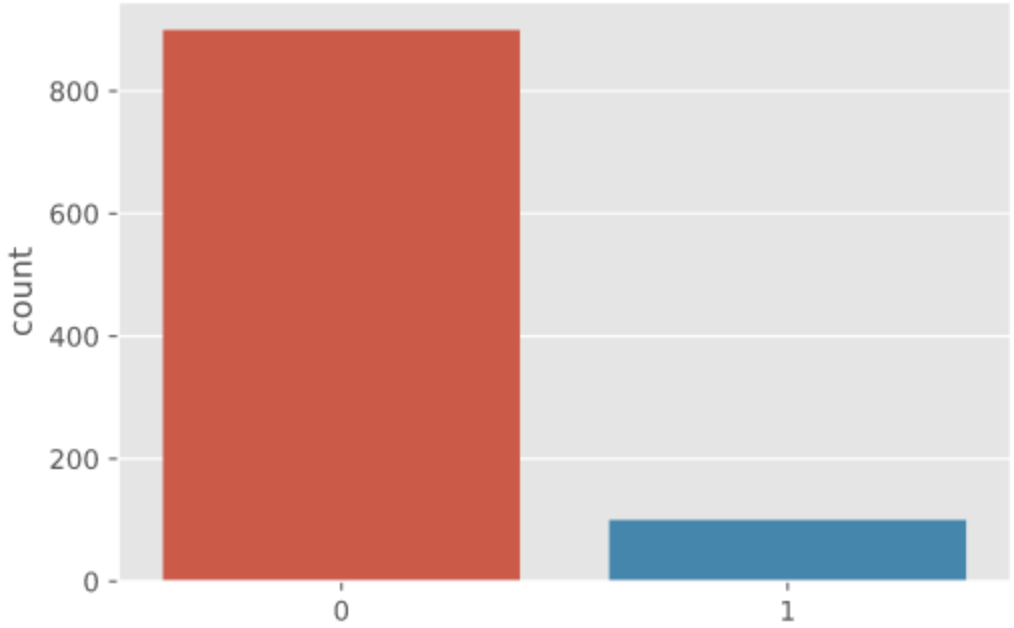
# Data: Imbalanced datasets

Dataset has more of one class than others.

This is very common, think of cancer diagnosis

Balance the train portion of the dataset only (not the test set)

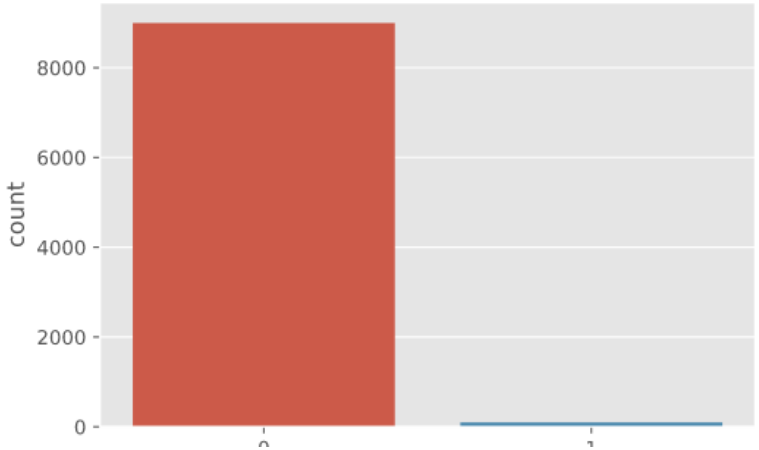
```
from sklearn.datasets import make_blobs
X, y = make_blobs(n_samples=[900,100], centers=None, cluster_std=[10.0, 2],random_state=22,n_features=2)
ax=sns.countplot(x=y)
```



# Data: Imbalanced datasets,

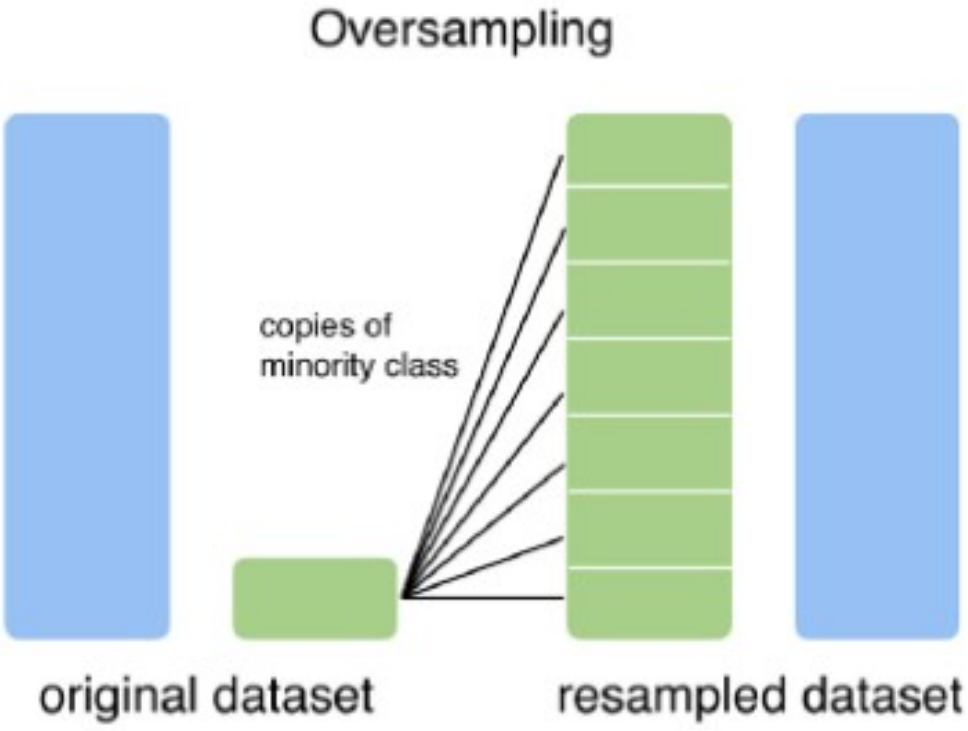
What if dataset is very imbalanced?
For fraud detection, 99.9% of transactions are legitimate.
Which means a fraud dataset consists of almost all legitimate transactions.
BTW for this case, if your evaluation metric is accuracy, a model may learn to always say its not fraud (99.9% accuracy!).

```
from sklearn.datasets import make_blobs
X, y = make_blobs(n_samples=[9000,100], centers=None, cluster_std=[10.0, 2],random_state=22,n_features=2)
ax=sns.countplot(x=y)
```



# Data: Imbalanced datasets, Guides

If you don't have a lot of data (<10,000 samples)
oversample the minority (train set only)




* https://machinelearningmastery.com/tactics-to-combat-imbalanced-classes-in-your-machine-learning-dataset/
https://medium.com/strands-tech-corner/unbalanced-datasets-what-to-do-144e0552d9cd/

# Data: Imbalanced datasets, Guides

Can also manipulate existing data.

- For images just crop sections to create new images
- Fiddle with the color (lighter,darker, redder, greener)
- Rotate image
- Others

Maybe the easiest (or hardest) of all;
Collect more data from the minority class

* https://machinelearningmastery.com/tactics-to-combat-imbalanced-classes-in-your-machine-learning-dataset/
https://medium.com/strands-tech-corner/unbalanced-datasets-what-to-do-144e0552d9cd/

# Data: Imbalanced datasets, Guides


**Balance this part only!**

The model will learn what distinguishes each class
It will not base decisions on relative class size

**DO NOT BALANCE THIS PART**

* https://machinelearningmastery.com/tactics-to-combat-imbalanced-classes-in-your-machine-learning-dataset/
https://medium.com/strands-tech-corner/unbalanced-datasets-what-to-do-144e0552d9cd/

# Summary

Three types of machine learning
We are going to do supervised learning
Calculate all metrics on train set only
Use these metrics to normalize test set(mean, std_dev)
Dealing with unbalanced datasets (balance train set only!)